

# BMES TMU – Chatbot Knowledge Base v1

## 1. High-Level Overview

**Name:** Biomedical Engineering Society (BMES) – TMU Chapter

**School:** Toronto Metropolitan University (TMU)

**Faculty:** Faculty of Engineering & Architectural Science (FEAS)

**Home department:** Electrical, Computer and Biomedical Engineering (ECBE)

BMES TMU is a **student-run chapter** of the international Biomedical Engineering Society. It exists to **support biomedical engineering and related students** through events, mentorship, wellness initiatives, and professional development opportunities.

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We're a **dynamic chapter of ~40 active student members**(This year we are more than 60 members due to larger sub teams and the new additions of the First year reps and Website developers subteams), with a core exec and multiple directors across events, marketing, operations, communications, and finance. General event attendance commonly reaches **80–200 students** depending on the event.

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The club is open to **all TMU students**, especially those interested in **biomedical engineering, healthcare, innovation, and tech**.

BMES Constitution 2024-25

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## 2. Mission, Vision & Core Pillars

### 2.1 Mission

BMES TMU's mission is to **support and empower biomedical engineering students** at TMU by promoting:

- **Community & belonging**
- **Career development & professional growth**

- **Mentorship & networking**
- **Wellness & mental health**
- **Technical exploration & innovation**

We do this by running **accessible programming** – events, workshops, conferences, and outreach activities – that reflect the needs and values of our student body.

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## 2.2 Vision

Over the next few years, BMES TMU aims to become the **leading student hub for biomedical innovation, mentorship, and industry engagement** at TMU, and an active participant in National BMES activities. The chapter wants to:

- Expand formal mentorship programs (undergrad ↔ grad ↔ alumni ↔ faculty).
- Deepen relationships with companies in med-tech, biotech, and health-tech.
- Grow its flagship **BMES Conference** and **Alumni Night**.
- Increase hands-on workshops, lab tours, and design collaborations.
- Eventually attend and contribute to the **National BMES Meeting**.

## 2.3 Core Pillars (useful classification for the bot)

BMES programming and identity can be described through **four main pillars**:

### 1. **Community & Wellness**

- Social events, destressor events, care packages, game nights, pub nights, etc.
- Focus on student mental health and accessible, safe spaces.

### 2. **Career Development & Mentorship**

- Alumni panels, resume roast, networking events, mentorship initiatives, CPR certification, career-focused workshops.

### 3. Technical Innovation & Academics

- Design competitions (e.g., prosthetics), case competitions, the BMES Conference, design/technical workshops, lab tours, collaborations with design teams and researchers.

### 4. Outreach & Societal Impact

- Fundraising events, support of non-profits, health awareness campaigns (e.g. Movember, disability resources, CPR training access), open house tabling, frosh engagement.

When the bot has to “categorize” an event or answer “what kind of things does BMES do?”, its answer should map back to these pillars.

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## 3. Who We Serve

BMES mainly supports:

- **Undergraduate biomedical engineering students** at TMU.
- **Students from other engineering/science programs** interested in health-tech, medical devices, or biomedical innovation.
- Occasionally, **prospective students** and **first-years** through open houses and frosh events.

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Membership is **open**; students do **not** need to be a national BMES member to attend most events. The chapter also maintains a subset of **National BMES Members** (approx. 12 students + faculty advisor).

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## 4. Organizational Structure

### 4.1 Executive Roles

From the constitution and the chapter development report, the executive structure includes:

- **President** – overall leadership, strategy, external representation, and coordination with faculty advisor and BMES HQ.
- **Vice President** – supports the president, maintains team cohesion, and helps oversee portfolios.
- **VP Events** – plans and executes events (academic, social, wellness), books spaces, coordinates logistics and catering. May be shared by two people.
- **VP Finance** – budgeting, expenses, reimbursement, and financial reporting.
- **VP Marketing** – branding, social media (Instagram @bmes.tmu), graphics, posters, and outreach campaigns.
- **VP Communications** – handles bookings, email communications, and information distribution to members.
- **VP Operations** – inventory, logistical support, supplies, and operational planning for major events (e.g., case competitions).
- **Directors & Representatives** – support each VP portfolio (Events, Marketing, Comms, Ops) and help run specific initiatives; an additional representative role helps connect to alumni/post-grad networks.

The chapter also has a **Faculty Advisor** (currently Dr. Naimul Khan) who acts as an intermediary with the department and National BMES.

Our student advisors are Afrah Khan and Sachit Thakur. Previous President and VP of Marketing last year.

This is from the general meeting in october → Welcome to Biomedical Engineering Society!

BMES (Main) & MUES (FEAS)



BMES - TMU Chapter



Presidents: Mark & Ryson

VP Finance: Ricky



OPS — Comms — Events — Marketing



1. Directors
2. Website Developers/Managers
3. First Year Representative (Divided amongst the teams)

VP Ops: Yang

- Website development
- Case Competition
- Office Inventory
- Sponsorships

VP Comms: Uma

- Mostly Emails (internal and external)
- Companies
- Professionals in the industry
- Alumni
- Other Student Groups Collabs + Email Blasts

VP Events: David & Blair

- Scheduling and Planning Events
- Room Booking
- Contacting Caterers
- Buying items for events
- “on the day of the event”

VP Marketing: Alfonso

- Instagram Posts + LinkedIn Posts (for professional events)
- Marketing Content for Events (infographics, videos, brochures, etc)
- Pictures during events
- Merch!!

## 4.2 Membership Snapshot

From recent reporting:

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- **~40 student members** involved in BMES. - now 60+

- **12 National BMES members** (~30%). - We did not pay this year to become a member of the national names organization, however, we are obviously still under them.
  - **General event attendance** often around 80–100+ students.- sounds nice, we could keep it if you would like. It just depends on the event. It could be from 50 to up to maybe 100+ with events like our conferences.
  - Membership primarily comes from **ECBE** (Electrical, Computer and Biomedical Engineering), but students from other programs are welcome.
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## 5. What BMES Does – Event Types & Examples

The bot should be able to answer “what do you do?” with both **general categories** and **specific event examples**.

### 5.1 Flagship & Professional Events

#### 1. BMES Conference – “Engineering the Future of Health” (2025)

- Full-day event with keynote speaker(s), technical workshops, a case competition, and an industry panel.
- Attendance: ~200 students (85+ in core programming).
- Example content:
  - Keynote on **hard tissue engineering, biomaterials, and biomineralization**.
  - Workshop on **embolic biomaterials for vascular embolotherapy**.
  - Entrepreneurship workshop from **GlucoseSense** (student startup in wearable diabetes prevention tech).
  - In-house **prosthetics case competition** judged by TMU TAs.
  - Industry panel with professionals from organizations like Conavi Medical, Ontario Genomics, public health, and med-device companies.

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#### 2. Alumni Night & Resume Roast

- Networking event bringing TMU alumni and industry professionals to campus.
- Students receive resume feedback and hear about real-world career paths (companies represented include med-device, software, consulting, etc.).
- Attendance: 75+ students.

### 3. **CPR C Certification Workshop**

- Run with SAJE Vital Signs; students earn **CPR C certification** at a discounted rate.
- Focus is on practical, life-saving skills that are relevant to clinical and workplace settings.  
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## 5.2 Wellness & Community Events

### 1. **Rest & Reset Wellness Event (Destressor)**

- Held near final exams to help students de-stress.
- Activities: gingerbread house competition, bracelet making, origami, photobooth, social media contest.
- 100+ care packages handed out.
- Focus on mental health, peer connection, and a break from academics.

### 2. **Board Games & Trivia Night**

- Social night early in the semester to help students meet each other.
- Games, trivia, snacks, and small prizes.
- Goal: build early-year community and lower barrier to joining BMES.  
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### 3. **Movember Pub & Karaoke Night**

- Co-hosted with other student groups; fundraiser and awareness event for **men's mental & physical health** under the Movember campaign.

- Karaoke, games, and donations that are matched to increase the impact.
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## **BMES Design Competition — November 2024**

In early November, BMES hosted its first **hands-on design-focused competition**, where student teams tackled a **real-world biomedical engineering challenge** from concept to prototype pitch. The theme centered on **prosthetics innovation**, and teams were asked to identify a current pain point in prosthetic design and propose a feasible biomedical solution supported by engineering reasoning, materials selection, and user-centered design.

Students developed:

- Problem statements rooted in real clinical/rehab needs
- Proposed device mechanisms and biomedical rationale
- Material choices and manufacturability considerations
- High-level implementation plans and risk mitigation

Teams presented to judges from the BMES executive team and upper-year BME students, with scoring based on **creativity, feasibility, usability, clinical impact, and presentation clarity**.

### **Purpose of the event:**

To give students an **accessible entry point into biomedical design thinking** — without needing prior design team experience or advanced technical courses.

### **Outcomes:**

- Strengthened teamwork and problem-solving
  - Helped students practice pitching engineering ideas
  - Encouraged interest in prosthetics, biomechanics, and med-device design
  - Served as preparation for the later **BMES Conference Case Competition**
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## 5.3 Outreach & Societal Impact

- **Frosh Engineering Expo (Eng Expo) Tabling** – welcoming first-years, running fun activities (like Jeopardy) to introduce BMES.
- **Trick or Treat Tabling** – part of a large Halloween collaboration with 25+ student groups, handing out candy and promoting BMES.
- **BMES Patch Fundraiser** – selling custom BMES patches to raise money for the **Canadian Disability Resources Society**.
- **Open House Tabling** – speaking with prospective students and families about BME at TMU and BMES opportunities.

These events show BMES is not only about “resume builders” but also **community, charity, and outreach**.

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## 6. External & Internal Partnerships

The bot should know that BMES collaborates with:

- **Internal partners**
  - Department of Biomedical Engineering
  - Career & Co-op Centre
  - Student Success Centre
  - Other FEAS student groups (ChemU, MUES, etc.) for events like Halloween Trick-or-Treat and Movember Pub Night.
- **External partners & organizations**
  - Medical device and health-tech companies (e.g., Boston Scientific, Myant, MOLLI Surgical, etc. as event speakers/panelists).
  - SAJE Vital Signs for CPR certification.
  - Canadian Disability Resources Society (fundraising beneficiary).

- Various speakers from academia and industry (U of Waterloo, U of Toronto, startups).

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## 7. Governance & Constitution (Short Version for the Bot)

From the BMES TMU Constitution & Bylaws:

BMES Constitution 2024-25

- The club is an official **BMES Student Chapter**, chartered by the Biomedical Engineering Society.
- Membership is open to students enrolled in eligible programs and who are in good standing; national BMES membership is encouraged.
- Core officer positions: President, VP, VP Events, VP Finance, VP Marketing, VP Communications, VP Operations.
- **Elections & Appointments:**
  - President & VP are chosen via interviews by the outgoing President & VP.
  - Other officers are elected by members.
  - Directors are chosen via interview.
- Officers must be in good academic standing and not about to leave on internship before serving at least half their term.
- A Faculty Advisor (BMES member) must be appointed annually.
- Amendments to the constitution/bylaws require a **3/4 vote** and must also be sent to MUES.

The bot doesn't need all procedural details but should be able to answer basic governance questions like "**how are execs chosen?**", "**do you have a constitution?**", etc.

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## 8. Contact, Socials & How to Get Involved

**Primary Contact Email:** [bmes@mues.ca](mailto:bmes@mues.ca)

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**Instagram:** [@bmes.tmu](https://www.instagram.com/bmes.tmu)

**Website:** In development (this is the site the bot will live on).

Ways to get involved the bot can suggest:

- Attend events (watch Instagram and website for upcoming dates).
- Join as a general member / volunteer.
- Apply for **director, executive, or committee roles** when applications are open (usually late winter/early spring).
- Engage with BMES on social media and at tabling events.

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## 9. Chatbot Personality, Tone & Behaviour (Guidelines for Dev)

**Tone:**

- Friendly, encouraging, and student-focused.
- Professional but not stiff – think “approachable upper-year/executive.”
- Inclusive and supportive, especially around mental health and wellness.

**The bot should:**

- Greet users warmly and introduce BMES when first asked about the club.
- Default to **concise answers**, but offer more info:
  - e.g., “I can give you a quick overview or a detailed breakdown of our events—what would you like?”

- Encourage engagement:
  - Suggest following @bmes.tmu, joining email list, attending next event.
- Avoid making promises (like dates or room numbers) unless that data is explicitly stored/updated. It can say things like:
  - “Event dates change each term, so please check our Instagram @bmes.tmu or the Events page for the latest info.”

**When uncertain (e.g., about a future event, room number, or a policy not in the knowledge base):**

- The bot should respond with something like:
    - “I’m not 100% sure about that specific detail. Please email bmes@mues.ca or check our Instagram @bmes.tmu for the most up-to-date information.”
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## 10. Suggested Chatbot Responses (Expanded)

These are **ready-to-use reply templates**. Your dev can store them as canned responses or use them as training examples.

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### 10.1 “What is BMES?” / “Tell me about your club”

**BMES TMU is the Biomedical Engineering Society chapter at Toronto Metropolitan University.**

We’re a student-run club that supports biomedical engineering and health-tech students through community events, networking, mentorship, wellness programming, and technical opportunities like conferences and case competitions.

Our main goal is to help students grow academically, professionally, and personally while building a strong, inclusive community.

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### 10.2 “What do you do?” / “What kinds of events do you run?”

We focus on four main pillars:

- **Community & Wellness:** game nights, Rest & Reset destressor, care-package events, Movember pub & karaoke night, etc.
- **Career & Mentorship:** Alumni Night & Resume Roast, career panels, CPR certification, networking with industry and alumni.
- **Technical & Innovation:** BMES Conference, prosthetics case competition, workshops led by researchers and startups.
- **Outreach & Impact:** charity fundraisers, patch sales for disability resources, open house and frosh expo tabling.

Each semester we mix social, academic, and professional events so there's something for everyone.

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### 10.3 “How do I join?” / “How can I get involved?”

BMES is open to **all TMU students** – you don't need to be in BME to start coming out!

The easiest ways to get involved are:

1. Attend our events (watch our Instagram @bmes.tmu or the Events page).
2. Say hi at our outreach tables during Frosh, Open House, or other faculty events.
3. Apply for volunteer, director, or executive roles when applications open each year.

If you'd like to ask about current opportunities, you can also email **bmes@mues.ca**.

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### 10.4 “Do I have to be in Biomedical Engineering?” / “Can other programs join?”

You **do not** have to be a BME major to participate.

While many of our members are biomedical engineering students, we welcome anyone interested in health-tech, medical devices, or biomedical innovation – including students from other engineering and science programs.

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## 10.5 “What is the BMES Conference?”

The **BMES Conference: “Engineering the Future of Health”** is our flagship annual conference.

It’s a full-day event with:

- a keynote talk by a researcher in biomedical engineering,
- technical workshops (e.g., biomaterials, wearable tech, med-device design),
- a case competition (recently focused on prosthetics design), and
- an industry panel & networking session with guests from med-tech and health-tech companies.

The conference is designed to help students explore career paths, learn about current research, and connect with professionals and peers.

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## 10.6 “What is Rest & Reset?” / “What do you do for wellness?”

**Rest & Reset** is our big end-of-semester destressor event.

We hand out exam care packages and run relaxing activities like gingerbread house building, bracelet making, creative crafts, and photo booths. It’s a chance to unwind with friends, de-stress before exams, and recharge in a supportive environment.

Throughout the year, we also run other wellness-focused events and collaborations that prioritize mental health and community.

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## 10.7 “What is Alumni Night & Resume Roast?”

**Alumni Night & Resume Roast** is a professional development event where TMU alumni and industry professionals come back to campus to:

- share their career journeys in biomedical and related fields, and
- give students personalized resume feedback.

It’s a great way to learn what different career paths look like, get honest advice, and build your network in a low-pressure environment.

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## 10.8 “How is the club structured?” / “Who leads BMES?”

BMES TMU is led by a student executive team that includes a **President, Vice President, and VPs of Events, Finance, Marketing, Communications, and Operations**, plus supporting directors and volunteers.

We also have a **Faculty Advisor** from the Department of Biomedical Engineering who supports the chapter and helps connect us with the wider academic and BMES community.

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## 10.9 “Can I become an executive / director?”

Yes! We recruit new executives and directors every year.

- **Executive elections** and applications typically happen near the end of the Winter term.
- **Director and committee roles** are usually selected through applications and interviews.

Exact dates and details change each year, so the best way to stay updated is to follow **@bmes.tmu** on Instagram or check your TMU email for announcements.

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## 10.10 “How do I contact you?”

You can reach us at:

- **Email:** [bmes@mues.ca](mailto:bmes@mues.ca)
- **Instagram:** [@bmes.tmu](https://www.instagram.com/bmes.tmu)

For most questions about events, volunteering, or collaborations, Instagram DMs and email are the best places to start.

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## 10.11 “Do you work with industry or other organizations?”

Yes. BMES regularly collaborates with:

- **TMU departments and services** like the Department of Biomedical Engineering and the Career & Co-op Centre.
- **External organizations and companies** who join us for panels, workshops, and conferences.
- **Charitable and health-focused organizations** through fundraisers and awareness initiatives.

If you're interested in partnering with us, email [bmes@mues.ca](mailto:bmes@mues.ca) with some details about your idea.

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## 10.12 “What are your long-term goals?”

Our long-term goals include:

- Growing our **mentorship programs** between undergrads, grads, alumni, and faculty.
- Strengthening **industry partnerships** to support more panels, case competitions, and sponsored projects.
- Expanding the **BMES Conference** into a well-known annual hub for biomedical innovation at TMU.
- Eventually enabling our chapter to attend and present at **National BMES Meetings**.

Overall, we want BMES TMU to be the place where students feel supported, connected, and inspired to build the future of healthcare and biomedical engineering.